

BATTA ADITYA YADAV

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SUMMARY

Applied AI Engineer with 5+ years of experience building production-grade AI/ML and agentic systems processing 10M+ events per day. Expert in Python, MCP, TypeScript, Voice Agents, and MLOps, with a proven track record of scaling enterprise AI, improving evaluation quality, optimizing latency, and delivering over \$11M+ in cost savings

WORK EXPERIENCE

ScitoVation, North Carolina, USA

Jan 2026– Present

AI Software Engineer – Founding Engineer (0 → 1) | Full time | Remote

- Drove 0→1 development of an AI-driven toxicology dossier system using Python, TypeScript, FastAPI, and MCPs, delivering **GCP**-production-based regulatory-grade reports in minutes instead of months for the scientific community
- Designed an orchestrated multi-agent architecture with tiered model routing and prompt caching, processing 335M+ tokens across 5,500+ LLM calls per day, replacing manual work and reducing costs by approximately 95% (**\$5M USD**)
- Extracted multimodal data from scientific PDFs, turning figures, tables, and prose into audit-ready Excel datasets and interactive charts, and improving latency from 30s → 14s by upgrading to Vision LLMs
- Fine-tuned LLMs using **QLoRA** and Opus for domain-specific use cases, defined model trade-offs, and hardened the evaluation pipeline by measuring faithfulness and recall using **MCP** tools & improved response quality by 93%
- Built evals using Snowflake's CortexAI with systematic scores (e.g. context relevance) to calibrate hybrid search and prompt strategies, reducing hallucinated outputs by tightening context alignment between retriever and LLM
- Built an end-to-end observability and provenance framework with **MLflow/PostgreSQL**, providing cost tracking and 100% traceability and reproducibility from every extracted value back to its original source
- Achieved 5× **RAG** latency reduction (250ms → 45ms) by introducing staged hybrid retrieval (metadata pre-filter → vector search → post-filter), improving **recall** to 95% via adaptive **chunking** and reducing irrelevant context by 75%
- Achieved research industry first AI vision that will extract exact data points from scientific figures with few-shot prompting design, and **reinforcement learning**, improving figure coordinates precision by 96%

Northeastern University & Massachusetts Institute of Technology, Boston, USA

Feb 2024– Jan 2025

Research Assistant- AI Engineer | Part time | Remote

- Designed a multilingual AI chatbot using **LangGraph** and **RAG**, achieving 78% response accuracy on NEU datasets
- Automated EKS deployments and CI/CD (GitHub Actions → EKS), standardizing reproducible release pipelines and reducing release time by ~85%, improving developer velocity and SDLC predictability
- Architected a **multi-agent** asynchronous workflow enabling parallel task execution and reasoning across AI agents
- Transformed unstructured data into optimized SQL schemas, enabling 99% query accuracy with **MLflow** tracking
- Developed end-to-end **MLOps** pipelines using **DAGS**, Airflow, and GitHub Actions for **CI/CD** automation
- Integrated Tensorflow for cosine similarity of research transcript metrics to automate data using pytest and Git
- Trained **AI agents** using Reinforcement Learning on **Hugging Face** transformers for monitoring & **Observability**
- Integrated Voice AI agents with Eleven Labs, Live Kit, and Twilio to enable natural, sub-500ms phone interactions

Johnson & Johnson, MA, USA

Jun 2024 –Dec 2024

Delivery Operations Supply Chain-AI Engineer Coop (0 → 1) | Full time | Hybrid

- Developed interactive **Power BI** dashboards tracking space performance for return inventory using **NoSQL MongoDB**
- Deployed an **LLMOps**-driven AI SQL chatbot integrating **BigQuery**, DAGs with **pydantic**, FastAPI and RESTful API for order tracking, and dashboard analysis on **Kubernetes** auto scaling **load balancer** with **Docker** image
- Designed bronze-silver-gold data pipeline for structured and unstructured MedTech vitals data (PDF) with **chunking**, **QA**, prompting and metadata strategies to optimize LLM-driven analysis for stakeholders and **Data version control**
- Conducted **statistical** hypothesis testing (ANOVA, t-tests, p-value analysis) with **time-series** trend and seasonality analysis to validate aisle space utilization, returns, and inventory performance, improving decision accuracy by 28%
- Optimized MedTech financial transactions with **VBA** and **Anthropic** File system; reducing cost risk analysis by 20%
- Applied **predictive modeling** for anomaly detection using ARIMA and Random Forest in Depuy Synthes instrument outliers, yielding **\$6M** CIP savings

Vitel Global, New Jersey

Aug 2021– July 2023

ML Scientist | Coop + Full time | Onsite

- Developed web applications using Node.JS, C++, JavaScript, and **React.js** on AWS EC2 improving efficiency by 47%
- Built **Python ETL** pipeline to optimize **Hive** queries for large-scale data extraction across 10+ TB of telecom datasets
- Applied advanced data mining and Statistical analysis on VoIP payment transactions using **pandas** and **scikit-learn**, identifying fraud and revenue leakage, and recovering \$0.3M+ in annualized revenue using **Random Forest**
- Designed facial-ID detectors using Multi-head **Attention**, CNN, **CUDA**, NVIDIA GPU and JAX during web-calls

- Performed **L2 regularization** and cross-validation to prevent overfitting and improve forecast reliability with **Hyperparameter** tuning and F1 score, AUC, Sensitivity, specificity and other key performance metrics by 14%
- Co-worked in **Monte Carlo Simulation** project to find revenue and budget outcomes

Stic Soft, Hyderabad

Dec 2020– July 2021

Full stack developer | Full time- University Program| Onsite

- Delivered 5+ full-stack web applications using **.NET**, **C#**, **PHP**, and **FastAPI**, improving backend performance and reducing API response time by 30%
- Built 15+ high-performance, responsive UI components using **React.js**, optimizing state management (**Redux/Context API**) and improving user engagement by 30% while reducing re-render time by 20%.
- Implemented and tested 20+ REST APIs using **Postman**, ensuring 99% API reliability and faster debugging cycles
- Configured and optimized reverse proxy (**Nginx**) and load balancing strategies, reducing server load by 40%, improving request throughput by 35%, and decreasing latency by 25% through efficient traffic distribution
- Collaborated in an **agile** environment to deliver high-quality features, demonstrating strong problem-solving skills and ownership in full-stack development projects
- Streamlined real-time data pipelines in **Snowflake** and **Apache Kafka**, increasing for high-volume streaming data

SELECTED PROJECTS

Open-Source Python Library: Unstructured data to Relational Database – QuerySUTRA 0.6.2 *Nov 2025*

- *launched a python library to parse pdfs with Agents, **Firecrawl**, **embeddings**, **semantic search**, **snorkel** and **FAISS DB***

Financial Stock Analytics: Multilingual Conversational Chatbot & Video Intelligence- MCP Tools *Aug 2025*

- *Build a **ReACT** chatbot using **NLP**, **langchain**, **RAG**, **FastAPI**, **MCP**, **anthropic** and **LLMs** for video summarization*

Deep Learning Prediction to find Intrinsically disordered proteins for drug discovery *May 2025*

- *Developed a **transformer-Pytorch** with **Meta's ESM-2** to classify 1,270 **DisProt** protein **IDRs**, achieving 81.7% accuracy*

Multi-Agentic AI Application: Rental Car Services Optimization (USA) - CrewAI *Feb 2025*

- *Built a **CrewAI** model using **Memory**, **GCP**, and real-time **Kayak** data for demand forecasting and pricing optimization*

Agent Firewall Web Scraper: Automated URL Multimodal Data – Firecrawl & BrowserBase *Jan 2025*

- *Web-scraped any URL into structured **JSON** using **Firecrawl** and **Gemini**, with **Playwright** as a fallback method*

TECHNICAL SKILLS

Programming & Tools : Python, MATLAB, Java, R, C++

Databases : PostgreSQL, Vector DB, Pinecone, Snowflake, Weaviate, Yaml, Airflow, Redis

Cloud : AWS EC2/ Lambda, GCP, Azure

ML & NLP : Decision Modelling, PC, Fuzzy words, NumPy, sentence-transformers, snorkel, **Tesseract OCR**

GenAI : MCP, CoT, ReAcT, LoRa, PEFT, Chroma DB, Langchain, Prompting, RAG chatbot

Monitoring : MLflow, Google cloud Monitoring, Langsmith

Deep Neural Networks : Transformers, BERT, GPT, GAN, CUDA, Kernel, RNN, LSTM

EDUCATION

Northeastern University, Boston, MA, USA

05/2025

Master of Science in Computer Science Engineering

GPA: 3.8

*Related courses: **Machine Learning Operations**, **Statistics**, **Neural Networks** and **Deep**, **Data Mining**, **Data Management** for **Analytics**, **Computational** and **Visualization**, **Foundations of Data Analytics**, **GenAI**, **Economic Decision Making***

CERTIFICATIONS

- [Career Essentials in Generative AI by Microsoft and LinkedIn](#)
- [ServiceNow IT Leadership Professional Certificate](#)
- [Google Data Analytics Capstone: Complete a Case Study](#)
- [McKinsey.org Forward Program](#)

AI/ML CONTRIBUTIONS

- Query Sutra Python Library – To convert unstructured data into relational databases - <https://lnkd.in/dMUrcbhb>
- Worlds first “AI detected coordinates on scientific graphs” – Huge Breakthrough in vision LLM’ (ScitoVation)